SCIENTIFIC AMERICAN

MATHEMATICAL RECREATIONS

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Source: *Scientific American*, Vol. 263, No. 1 (JULY 1990), pp. 118-121

Published by: Scientific American, a division of Nature America, Inc.

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MATHEMATICAL RECREATIONS

An odd journey along even roads leads to home in Golygon City

by A. K. Dewdney

"A journey of a thousand miles must begin with a single step."

—LAO-TZU,
The Way of Lao-tzu

Allow me to start you on a journey in Golygon City. You can take a similar trip in New York, Kyoto or almost any large city whose streets form a grid of squares. Here are your directions. Stroll down a city block, and at the end turn left or right. Walk two more blocks, turn left or right, then walk another three blocks, turn once more and so on. Each time you turn, you must walk straight one block farther than before. If after a number of turns you arrive at your starting point, you have traced a golygon. If you do not need the physical exercise, you can easily simulate the journey by moving a pencil along a piece of graph paper with a square grid. If you become lost, you may refer to the map below.

A golygon consists of straight-line segments that have lengths (measured in miles, meters or whatever unit you prefer) of one, two, three and so on, up to some finite number. Every segment connects at a right angle to the segment that is one unit larger—except the longest segment, which meets the shortest segment at a right angle. Golygons are more than just a geometric curiosity. They have inspired some delightful puzzles as well as some intriguing problems for research. What better way to develop insight into the research process than to take a recreational journey?

A map of Golygon City

It seems that golygons were first conceived by Lee Sallows, a redoubtable engineer of the Catholic University Nijmegen in the Netherlands. I last featured Sallows's work in October, 1984, when I described his search for pangrams, sentences containing each letter of the alphabet. Since then Sallows has invented many new recreations but none so engaging, as far as this department is concerned, as golygons.

In the fall of 1988 Sallows began his search for golygons. It did not take him long to find an eight-sided golygon. Yet he could find no such objects that had fewer than eight sides. Nor did he discover a golygon with nine sides, nor 10, nor 11,...until at last he stumbled on a 16-sided golygon.

Wondering whether he had missed any, Sallows wrote a computer program to automate the search. It spewed out no less than 28 different 16-sided golygons [*see illustration on opposite page*] before moving on to higher orders. The program did not find golygons with from 17 up to 23 sides, but it generated numerous 24-sided golygons—2,108 to be exact.

By then Sallows had a hunch that the number of sides in a golygon must be a multiple of eight. Yet his program, which was already laboring in the 20's, was unequal to the task of discovering any 32-sided golygons. Frustrated, Sallows appealed to Martin Gardner, the supreme authority in matters of recreational mathematics. Could Gardner prove that the number of sides in a golygon must be a multiple of eight?

We can creep up on Gardner's proof by starting with the simpler task of proving that the number of sides must be a multiple of two. To explain, allow me to take you back to Golygon City. We start our journey by walking one block to the north. (I choose north only for convenience.) Hence, the first side of the golygon will run north for one block. Next we can turn either left or right, that is, west or east. Therefore, the second side of the golygon will run either two blocks to the west or two blocks to the east. It follows that all odd sides in the golygon (the first, third, fifth, etc.) are an odd number of blocks long and run north or south; all even sides (the second, fourth, sixth, etc.) are an even number of blocks long and run east or west. Because the last side of the golygon meets the first at right angles, the last side must run east or west. Therefore, the last block is an even side of the golygon, and the number of sides in a golygon must be a multiple of two.

To show that the number of sides must be a multiple of four, we begin by

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calculating the total distance we traveled to the north of the starting point. To do this, we simply add the number of blocks we walked to the north and subtract the number of blocks we traveled to the south. (A net negative distance to the north simply means that we are south of the starting point.)

Because all north and south sides are an odd number of blocks long, we are essentially adding and subtracting consecutive odd numbers. It takes only a little tinkering to convince ourselves that the result is always even when we add or subtract an even number of odd lengths—for example, $1+3-5+7=6$. By the same token, an odd number of odd lengths always gives an odd sum. Therefore, the distance walked north is an even number of blocks if and only if we have walked along an even number of north and south sides.

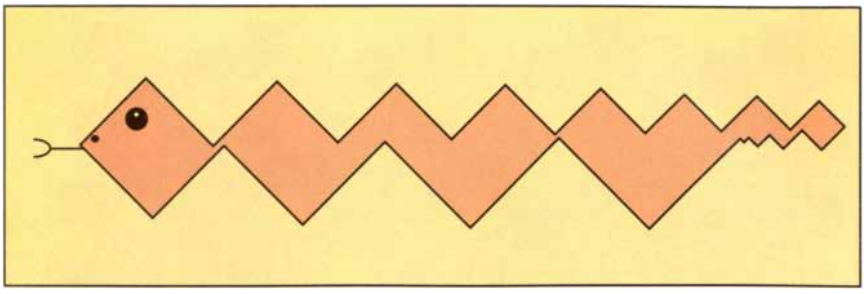
Now if we walk from the starting point around Golygon City and return, the distance to the north of the starting point equals zero. Because zero is an even number, the golygon must have an even number of north and south sides. The total number of sides is twice the number of north and south sides, because for every north or south side there must be an east or west side. Therefore, the number of sides in a golygon is a multiple of four.

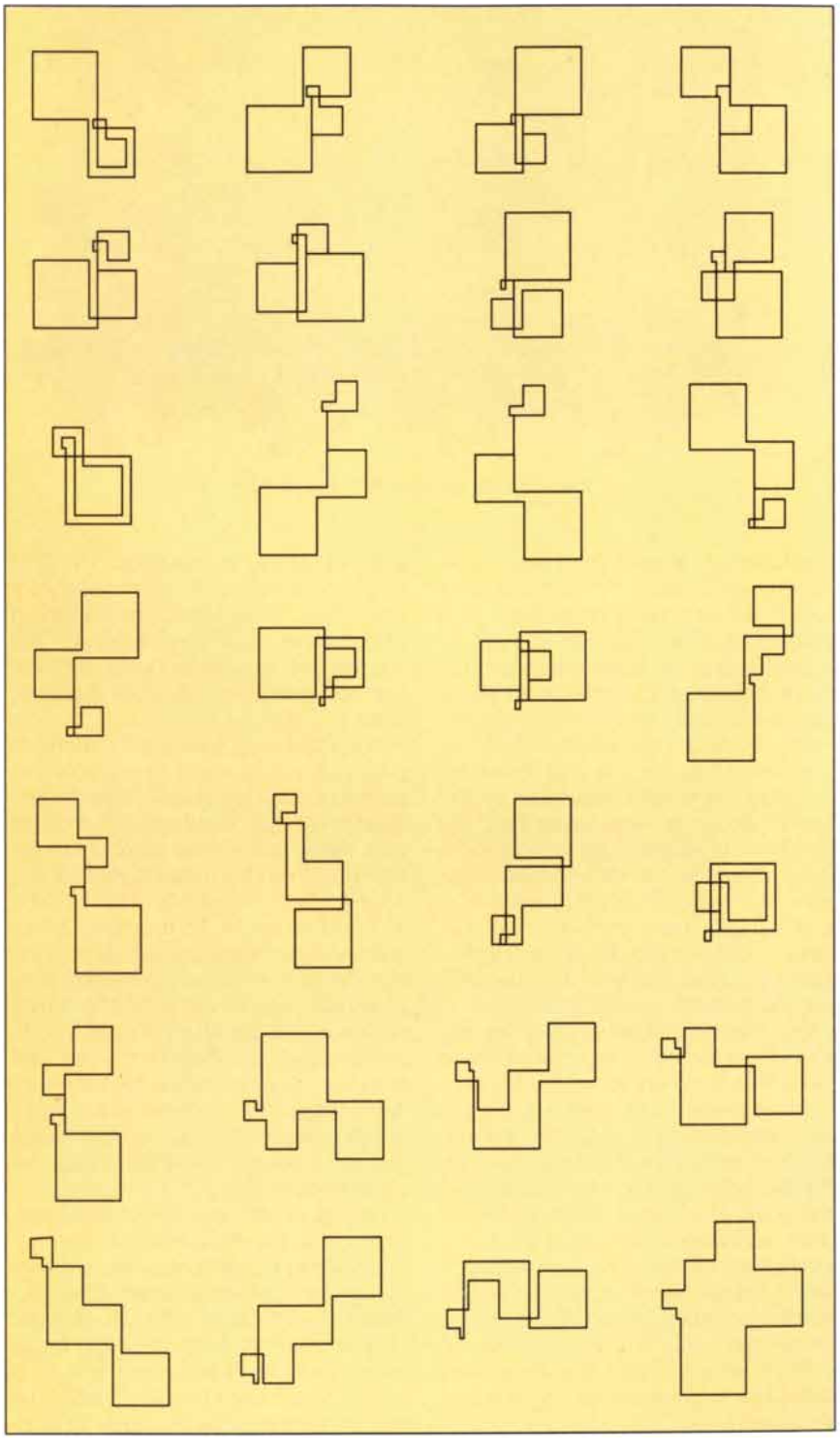
How on earth did Gardner show that the number of sides must be a multiple of eight? Let us accompany the master on his journey. We already know that the number of north and south sides in the golygon is even. We also know that the distance traveled to the north must equal the distance traveled to the south. Therefore, the distance north must equal half the total distance that we have walked in both directions, north and south. Hence, the total distance is equal to the sum of an even number of consecutive odd numbers. For instance,

$$1+3=4$$
$$1+3+5+7=16$$
$$1+3+5+7+9+11=36$$
$$1+3+5+7+9+11+13+15=64$$

The result always seems to be a multiple of four.

If the total distance north and south is equal to a multiple of four, then half that distance must be a multiple of two. Therefore, the distance we traveled north must be a multiple of two, and the number of north sides must also be a multiple of two because all north sides are an odd number of blocks long. It follows that the number of south sides is a multiple of two and that the total number of north and





One golygon snake that has 32 sides and all 28 golygons that have 16 sides

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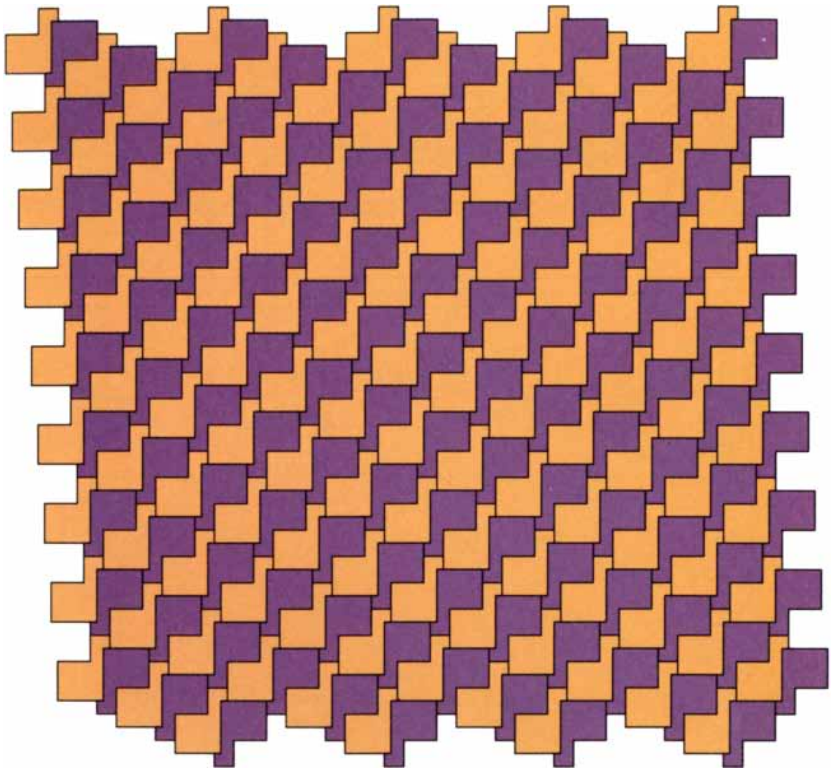

The smallest golygon tiles the plane

south sides is a multiple of four. Finally, after we include the east and west sides, the total number of sides in a golygon must be a multiple of eight.

Gardner's proof may seem rather peculiar because odd numbers played a major role and even numbers hardly entered into the discussion. The same conclusion can be reached, however, by making a similar argument on behalf of the even lengths. In fact, the two kinds of lengths play independent roles in the construction of any golygon. One may obtain, for example, a new golygon from two others of the same size by using the even lengths from the first golygon and the odd lengths from the second.

Our recreational path has so far taken us from the question of existence to the question of conditions on that existence; the number of sides in a golygon must definitely be a multiple of eight. For convenience I will call this theorem the $8k$ condition for two-dimensional golygons. It is what mathematicians call a necessary condition. If a golygon exists, the number of its sides is necessarily a multiple of eight. But is this condition sufficient? In other words, if we merely ask that the number of sides, n, be a multiple of eight, is that sufficient to guarantee the existence of a golygon?

We can find a golygon that has eight times any number of sides by applying a set of simple instructions. The first step is to decide how large a value of n one wants. To illustrate the method, I will choose a 16-sided golygon. The second step is to write down consecutive numbers from 1 to n, in this case, from 1 to 16.

The third step would have us place plus and minus signs in front of the numbers. The first quarter and the last quarter of the numbers will receive plus signs, and all the numbers in between will receive minus signs: +1, +2, +3, +4, -5, -6, -7, -8, -9, -10, -11, -12, +13, +14, +15, +16. To translate this sequence of numbers into a tour of Golygon City, you need only know that positive odd numbers run to the north, negative odd numbers to the south, positive even numbers to the east and negative even numbers to the west. The golygon that corresponds to these numbers looks like a snake. A 32-sided "snake" is shown in the illustration on the preceding page.

Having settled the question of sufficiency of the $8k$ condition, Gardner and Sallows began to wonder just how many golygons should exist. They had found one of order eight, 28 of order 16 and 2,108 of order 24. How many would there be of higher orders? Such questions are most frequently asked by mathematicians who specialize in combinatorics, the study of discrete mathematical objects such as sets and graphs [see "Ramsey Theory," by Ronald L. Graham and Joel H. Spencer, page 112]. Gardner decided to check with some experts. He wrote letters to computer scientist Donald E. Knuth of Stanford University and to mathematician Richard K. Guy of the University of Calgary in Alberta. Could either of them assist in the matter of enumerating golygons?

Before long, Knuth had written a computer program that counted all the golygons up to 64 sides. The total number of 64-sided golygons, for instance, is 127,674,038,970,623. Unfortunately, Knuth's program became unwieldy at larger sizes.

Meanwhile Guy was able to develop a formula that expressed the approximate number of golygons for each possible value of k, where k equals the number of sides divided by eight. The formula consists of a fraction, the principal parts of which are 2 raised to the power $8k-6$ in the numerator and k cubed in the denominator. Because, as k increases, the exponential term increases much more rapidly than the cubed term, the growth in the number of golygons is basically exponential. Guy's formula has the further important property of being asymptotic, that is, it gives a value that, expressed as a percentage of the true value, comes increasingly close to 100 percent as k gets larger and larger.

Sallows, Gardner, Knuth and Guy decided to report their various results in a paper. The word "golygon" does not appear in the title. Instead it is called "Serial Isogons of 90 Degrees," a name that rings more solemnly than "golygons." The reference to 90 degrees implies that other kinds of golygons might be assembled. Indeed, some of the authors have looked at golygons in which every angle is not 90 degrees but 60 or 120 degrees. To construct such golygons, readers may need hexagonal-grid paper, which can be found in most graphics-supply stores.

Would this have been a good place to stop? Research (like recreation) never stops. Often, as research in one question progresses, other questions suggest themselves. Sometimes it is very hard to decide just what question to work on.

Research problems are no sooner solved than the recreationist in us thinks of new avenues of inquiry: Can we find prime-sided golygons? The lengths of consecutive sides here increase by the sequence of odd primes: 1, 3, 5, 7, 11, 13 and so on.

Another offshoot of golygons deserves mention. Readers are urged to look at the eight-sided golygon once again. It is unique, the only member of

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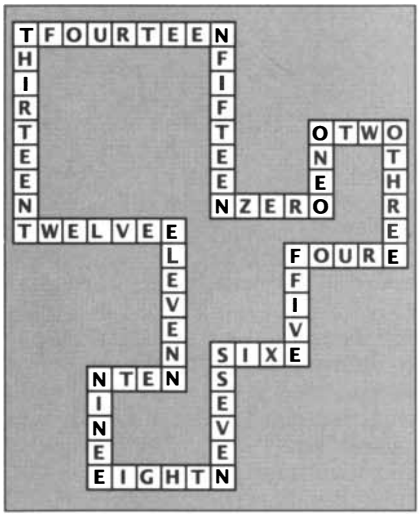
its class. Eight-sided golygons will fit together very nicely to make tiling patterns. They will, as the professionals say, "tile the plane." The illustration on the opposite page shows an attractive tiling.

Properly speaking, the region bounded by an eight-sided golygon should be called a polyomino. Not all golygons produce polyominoes because some golygons intersect themselves in their meandering march around the plane. In fact, polyominid golygons (a formidable expression by which readers can lose friends and influence people) are probably increasingly rare in relative terms as *n* gets large. Only four of the 29 golygons shown on page 119 are the boundaries of polyominoes. None of these appears to tile the plane.

But in the matter of tiling, Sallows has a challenge for readers: suppose the eight-sided golygon represents the boundary of one's kitchen floor. Given 13 L-shaped tiles, can readers cover the kitchen floor exactly with these eccentric tiles? For the purpose of this question, the floor may be thought of as divided into 52 squares, like the squares of the special paper on which this enquiry started. Each square is one step on a side. The L-shaped tiles consist of three squares in a row and an additional square comprising the foot of the L. Anyone who finds the task impossible must, of course, prove it to be so.

I close with some final words from Sallows, the engineer who started all this. The "words" are ZERO, ONE, TWO and so on, up to FIFTEEN. They can be arranged in a special golygon all their own in which the letters of these words determine the sides of what can only be called a logolygon [*see illustration below*].



A logolygon created by Sallow

Reader Response

The Emperor's New Mind *and wireworld*

The Pandora's box of minds, machines and metaphysics, which I opened last December, gave me a ringside seat to a battle between titans. In one corner stood Roger Penrose, whose recent book, *The Emperor's New Mind,* has given new heart to those who feel that the human brain is more than a computer. In the opposing corner were members of the artificial-intelligence and computing community who find nothing particularly new or incisive in the book.

The philosopher John R. Searle was called to defend Penrose [see "Is the Brain's Mind a Computer Program?" SCIENTIFIC AMERICAN, January]. Searle put forward the "Chinese room" argument: a computer program that can read and write Chinese does not understand Chinese anymore than a person who imitates the action of the program.

Among the champions of the artificial-intelligence community was Hans P. Moravec, a robotics expert at Carnegie-Mellon University, who sent me an open letter that he had mailed to Penrose. Moravec, whose gung ho attitude is abundantly evident in his book, *Mind Children,* calls Penrose's argument "wildly wrongheaded."

Would Penrose grant the same intelligence to a certain robot as he would to the dog that looks up at him with soft brown eyes and whines? The robot Moravec has in mind is at present more imaginary than real. It says, "*Please,* Roger, it bothers me that you don't think of me as a real person. What can I do to convince you? I am aware of you, and I am aware of myself. And I tell you, your reaction is almost unbearable." The robot's reaction is the result of a special program designed to convince the master that some caring is in order—perhaps like the whining of a dog.

I can only begin to sample the many readers who responded to this column. Jef Raskin of Pacifica, Calif., states that Searle's Chinese room argument is "tantamount to defining intelligence by stating, 'Intelligence is a nonalgorithmic process that…'" Paul Preuss, a computer aficionado and noted science-fiction writer from San Francisco, claims that the Chinese room is not a true thought experiment. It should be performed, Preuss says, "by Dr. Searle himself. If he succeeds in convincing a Chinese speaker that he, inside the room, really can read and write Chinese, he should then report on his subjective feelings as to whether or not he has learned Chinese in the process."

The famous incompleteness theorem of German mathematician Kurt Gödel is often cited as evidence that there are thoughts beyond the reach of algorithms. My query as to whether the theorem enjoys extra-algorithmic status brought a hail of "nays" from a variety of seeming experts, including Greg Ray, who studies the logic and methodology of science at the University of California at Berkeley, and from Robert Wilensky, director of the Berkeley Artificial Intelligence Project. Peter Kugel of the department of computer science at Boston College opined that Gödel's proof is itself an algorithm.

The final "Computer Recreations" column, appearing this past January, featured two cellular-automaton software packages created by Brian Silverman of Montreal and by Rudy Rucker of Los Gatos, Calif.

Wireworld, a two-dimensional cellular space in which home experimenters can set up simple computing circuits, inspired a number of readers to experiment on their own. Nyles Heise of Houston, Tex., shared Silverman's enthusiasm for simulated circuits to the point of writing a program that duplicated many features of wireworld. Written in the compact language APL in a single evening, the program enabled Heise to construct memory elements that surpassed my own in speed. Using wireworld, I constructed a circular path that would hold a single live pulse that could be sampled every 13 ticks of the cellular clock. Heise's most efficient memory element has a cycle time of only four ticks. To build such a speedy device, however, he had to invent a new way to transmit pulses along wires.

Other creative souls included Oliver Springauf of Würzburg, West Germany, and Davide P. Cervone of Providence, R.I. Following my more conservative technology, Springauf succeeded in improving my memory loop from 13 to 12 ticks. Cervone built an amazing variety of devices, including a four-bit adder. The layouts sent in by all readers looked like nothing on earth, so alien is the notion of cellular computing to the continuous physics employed by actual computing microcomponents.

FURTHER READING

SERIAL ISOGONS OF 90 DEGREES. Lee Sallows, Martin Gardner, Richard Guy and Donald Knuth in *Mathematics Magazine*. Edited by G. L. Alexanderson (in press).

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